

SQL Queries Used for Airbnb Market Analysis (New York City)

1. Determine the total Airbnb listings and unique hosts using the cleaned listings dataset.

SQL Code –

```
1      # Total Listings and Unique Hosts
2
3      • Select
4          count(*) as total_listings,
5          count(distinct host_id) as unique_hosts
6      from listings_cleaned
```

Output –

	total_listings	unique_hosts
▶	3348	2929

2. Analyze the percentage share, number of listings, and average price for each room type within each neighborhood group.

SQL Code –

```
# Compare % share room types within each neighbourhood group
Select
    neighbourhood_group_cleaned,
    room_type,
    count(*) as num_listings,
    round(count(*) * 100.0 / sum(count(*)) over (partition by neighbourhood_group_cleaned),2) as pct_share,
    Round(avg(price),2) as avg_price
from listings_cleaned
group by neighbourhood_group_cleaned, room_type
order by neighbourhood_group_cleaned, pct_share desc
limit 20;
```

Output –

	neighbourhood_group_cleaned	room_type	num_listings	pct_share	avg_price
▶	Bronx	Private room	35	64.81	59.66
	Bronx	Entire home/apt	18	33.33	116.94
	Bronx	Shared room	1	1.85	55.00
	Brooklyn	Entire home/apt	966	64.14	184.10
	Brooklyn	Private room	524	34.79	87.63
	Brooklyn	Shared room	11	0.73	57.36
	Brooklyn	Hotel room	5	0.33	204.40
	Manhattan	Entire home/apt	1026	67.68	266.10
	Manhattan	Private room	471	31.07	111.81

Insights –

- In Brooklyn and Manhattan, "Entire home/apt" is the most common room type, making up over 64% and 67% of listings in those areas, and commanding the highest average prices.
- Private rooms are the second most common in all shown neighborhoods, while shared rooms and hotel rooms have very low shares across the board.

3. Find the average price for each Airbnb room type in the dataset.

SQL Code –

```
1  # Average Price by room Type
2  •  select
3      room_type,
4      round(avg(price),2) as avg_price
5  from listings_cleaned
6  group by room_type
7  order by avg_price desc
8
```

Output –

	room_type	avg_price
►	Entire home/apt	220.99
	Hotel room	204.40
	Private room	94.36
	Shared room	88.65

Insights –

- "Entire home/apt" has the highest average price at 220.99, followed by hotel rooms at 204.40.
- Private rooms (94.36) and shared rooms (88.65) are much more affordable than the other options.

4. Identify the neighborhoods with the highest number of Airbnb listings.

SQL Code –

```
1  # Top 30 Neighbourhood by Number of Listings
2  •  Select
3      neighbourhood_cleansed,
4      count(*) as num_listings
5  from listings_cleaned
6  group by neighbourhood_cleansed
7  order by num_listings desc
8  limit 30;
```

Output –

	neighbourhood_cleansed	num_listings
▶	Williamsburg	291
	Bedford-Stuyvesant	247
	Harlem	234
	Upper West Side	168
	East Village	155
	Greenpoint	132
	Crown Heights	131
	Upper East Side	123
	Chelsea	96

Insights –

- Williamsburg, Bedford-Stuyvesant, and Harlem are the top neighborhoods by listing count, with 291, 247, and 234 listings respectively.
- The distribution shows that several distinct neighborhoods have strong Airbnb activity, not just one central area.

5. Compare average price and review score for Airbnb listings across neighborhood groups.

SQL Code –

```
1  # Location wise Avg price and review Score
2  •  Select
3      neighbourhood_group_cleansed,
4      Round(Avg(price),2) as avg_price,
5      Round(Avg(review_scores_rating),2) as avg_review_score
6  from listings_cleaned
7  group by neighbourhood_group_cleansed
8  order by avg_price asc
9  limit 5;
```

Output –

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	neighbourhood_group_cleaned	avg_price	avg_review_score			
▶	Bronx	78.67	4.71			
	Staten Island	92.57	4.72			
	Queens	102.23	4.64			
	Brooklyn	149.67	4.68			
	Manhattan	216.16	4.66			

Insights –

- Bronx has the lowest average price at 78.67, while Manhattan is the highest at 216.16.
- Despite lower prices, neighborhoods like Bronx and Staten Island have the highest average review scores.

6. Analyze the average yearly availability of each room type by neighborhood group.

SQL Code –

```
1      # AVerage Avalability (365 days)
2
3      •  SELECT
4          neighbourhood_group_cleaned,
5          room_type,
6          ROUND(AVG(availability_365), 2) AS avg_availability
7      FROM listings_cleaned
8      GROUP BY neighbourhood_group_cleaned, room_type
9      ORDER BY neighbourhood_group_cleaned, avg_availability DESC
10     limit 10;
```

Output –

	neighbourhood_group_cleaned	room_type	avg_availability
▶	Bronx	Shared room	365.00
	Bronx	Private room	236.17
	Bronx	Entire home/apt	146.94
	Brooklyn	Hotel room	281.80
	Brooklyn	Shared room	147.09
	Brooklyn	Entire home/apt	133.70
	Brooklyn	Private room	130.81
	Manhattan	Shared room	187.05
	Manhattan	Private room	151.44

Insights –

- Shared rooms in the Bronx have the highest average availability (365 days), indicating little to no bookings.
- Hotel rooms in Brooklyn and private rooms in the Bronx also show high availability, suggesting these categories may have lower occupancy rates.

7. Compare the average listing price versus reported average price for each neighborhood group, ranking neighborhoods by the largest price difference.

SQL Code –

```
1  # Comparison of the Average listing price for each Area
2
3  • WITH price_comparison AS (
4      SELECT
5          l.neighbourhood_group_cleaned,
6          ROUND(AVG(l.price), 2) AS listings_avg_price,
7          ROUND(AVG(n.avgp), 2) AS reported_avg_price,
8          ROUND(ABS(AVG(l.price) - AVG(n.avgp)), 2) AS price_difference
9      FROM listings_cleaned l
10     JOIN neighbourhoods n
11         ON l.neighbourhood_group_cleaned = n.neighbourhood_group_cleaned
12     GROUP BY l.neighbourhood_group_cleaned
13 )
14 SELECT
15     neighbourhood_group_cleaned,
16     listings_avg_price,
17     reported_avg_price,
18     price_difference,
19     RANK() OVER (ORDER BY price_difference DESC) AS rank_by_diff
20 FROM price_comparison
21 ORDER BY price_difference DESC
22 LIMIT 5
23 ;
24
```

Output –

Result Grid Filter Rows: Export: Wrap Cell Content:					
	neighbourhood_group_cleaned	listings_avg_price	reported_avg_price	price_difference	rank_by_diff
►	Queens	102.23	118.7	16.46	1
	Bronx	78.67	94.82	16.16	2
	Staten Island	92.57	106.17	13.6	3
	Manhattan	216.16	205.77	10.4	4
	Brooklyn	149.67	141.67	8.01	5

Insights –

- Queens and Bronx have the biggest differences between listing and reported average prices, at 16.46 and 16.16, indicating potential pricing inconsistencies or data mismatches.
- Brooklyn shows the smallest price gap (8.01), suggesting more consistent pricing data between sources.

8. Analyze the correlation between price and availability for properties managed by individual hosts.

SQL Code –

```
1  # correlation Between price and host property
2  SELECT
3      host_name,
4      ROUND(
5          (AVG(price * availability_365) - AVG(price) * AVG(availability_365)) /
6          (STDDEV(price) * STDDEV(availability_365)), 2
7      ) AS price_availability_corr,
8      CASE
9          WHEN ROUND(
10             (AVG(price * availability_365) - AVG(price) * AVG(availability_365)) /
11             (STDDEV(price) * STDDEV(availability_365)), 2
12         ) > 0.3 THEN 'Positive correlation (Higher availability → Higher price)'
13         WHEN ROUND(
14             (AVG(price * availability_365) - AVG(price) * AVG(availability_365)) /
15             (STDDEV(price) * STDDEV(availability_365)), 2
16         ) < -0.3 THEN 'Negative correlation (Lower availability → Higher price)'
17         ELSE 'Weak/No correlation'
18     END AS correlation_meaning
19 FROM listings_cleaned
20 WHERE price IS NOT NULL
21     AND availability_365 IS NOT NULL
22     AND host_name IS NOT NULL
23 GROUP BY host_name
24 HAVING COUNT(*) > 1
25     and price_availability_corr is not null
26 ORDER BY price_availability_corr ASC
27 limit 5;
```

Output –

Result Grid			
Filter Rows:			
Export: Wrap Cell Content: Fetch rows:			
	host_name	price_availability_corr	correlation_meaning
▶	Garon	-1	Negative correlation (Lower availability → Highe...
	Cyn	-1	Negative correlation (Lower availability → Highe...
	Sing	-1	Negative correlation (Lower availability → Highe...
	Teri	-1	Negative correlation (Lower availability → Highe...
	JT And Tiziana	-1	Negative correlation (Lower availability → Highe...

Insights –

- All displayed hosts have a strong negative correlation (-1), meaning higher prices are associated with lower availability.
- This suggests that as the price increases, those listings tend to be booked more frequently or consistently.

Conclusion

- Airbnb listing prices and availability vary significantly by neighborhood and room type, with entire homes having the highest average prices.
- Popular neighborhoods like Williamsburg and Bedford-Stuyvesant have the most listings, while prices can differ notably between listing data and reported averages.
- There is a strong negative correlation between price and availability, indicating pricier listings tend to be booked more frequently.
- These insights provide strategic value for hosts to optimize pricing and availability and help guests make better-informed accommodation choices.